Highly accurate protein structure prediction with AlphaFold

Jumper et al., *Nature* 596 (2021) [JEP+21]

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`github.com/BenBullinger/ML4Genomics-Presentation-AF2`

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

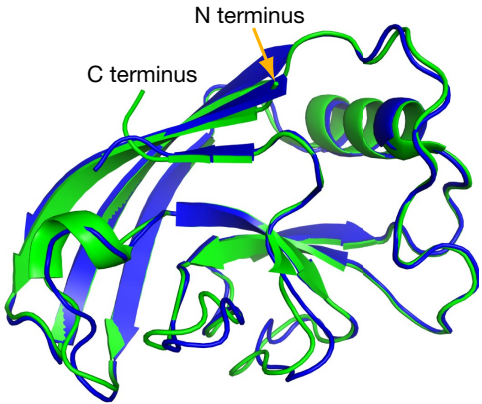
Protein Structure Prediction

SIFSYITESTGTPS...

Amino Acid Residues = Amino Acids after being incorporated into protein

The network predicts the 3D position of heavy atoms

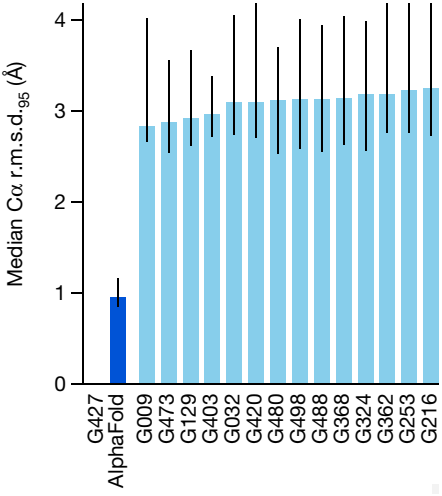
Heavy atoms = every atom except hydrogen



Experiment and prediction

[JEP+21] Fig. 1b

The Benchmark: CASP14



CASP14: Critical Assessment of protein Structure Prediction

May-July 2020

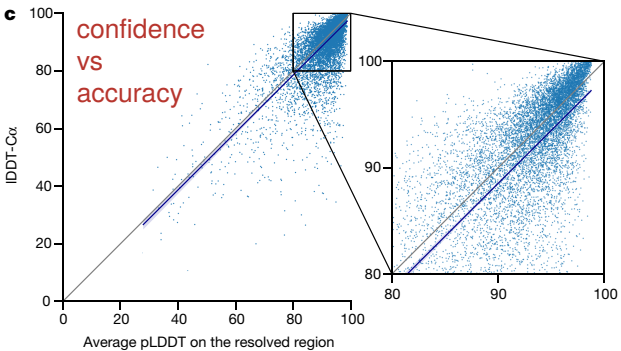
Blind test: experimental structures are not revealed until the end of experiment.

Å (angstrom) = unit equal to 10^{-10} meters.

Smaller RMSD means a closer structural match

[JEP+21], Fig. 1a, The performance of AlphaFold on the CASP14 dataset (n = 87 protein domains) relative to the top-15 entries (out of 146 entries)

Predicted Confidence



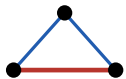
*(p)*LDDT: *(Predicted)* Local Distance Difference test

Local accuracy score calculated per residue, averaged over protein; 0 to 100.

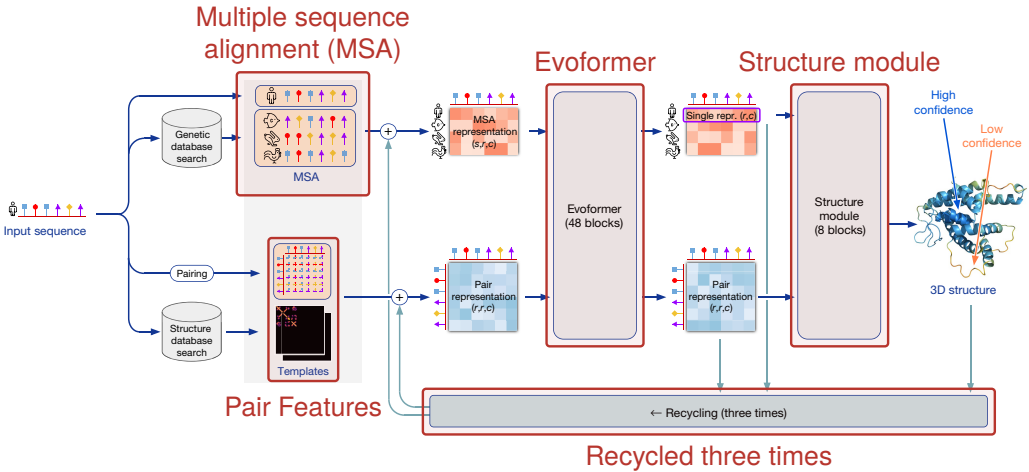
pLDDT is accurate as model's confidence score

[JEP+21], Fig. 2c, Confidence score compared to the true accuracy on chains

Design Constraints

	Constraint	AlphaFold 2
1 	Triangle inequality on distances	Pair updates over triangles
2 	Rotation and translation invariance	Invariant attention (IPA block)
3 	Chirality (mirror images differ)	FAPE loss over local frames

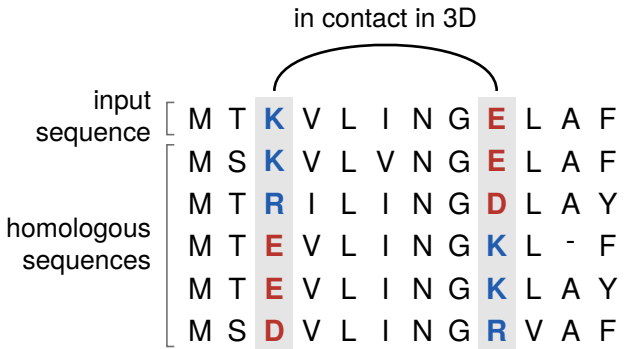
Network Architecture



Recycling: execute network multiple times on the same input.

[JEP+21], Fig. 1e

Multiple Sequence Alignment (MSA)

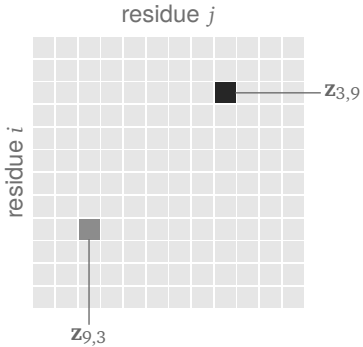


Co-evolution: residues in contact change together across homologous proteins.

Homologues = evolutionary related

Illustration after [MCS+11]

Pair Representation



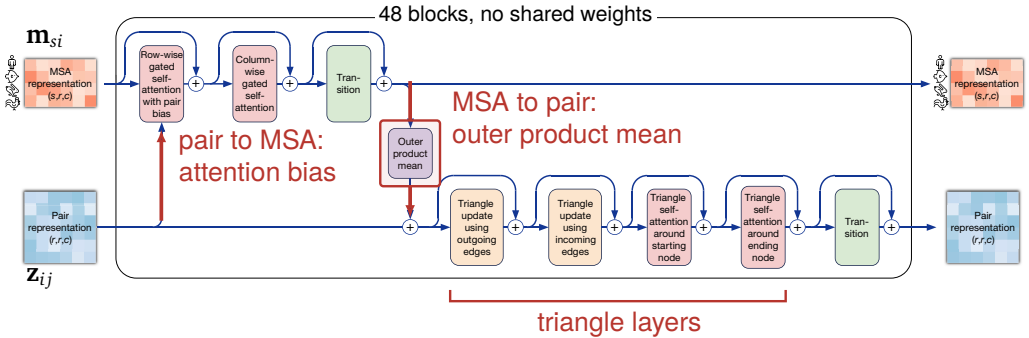
Pair representation $\mathbf{z}_{ij}$: one vector per ordered pair of residues. $\mathbf{z}_{ij}$ and $\mathbf{z}_{ji}$ can differ.

Start: the two amino acids, their offset $i - j$, and their distance in *templates*, known structures of related proteins.

An auxiliary loss trains $\mathbf{z}_{ij} + \mathbf{z}_{ji}$ to predict the distance of i and j.

[JEP+21], p. 585, Fig. 3b. SI 1.2.9, 1.5, Algorithms 2 to 4. Distance loss: SI 1.9.8

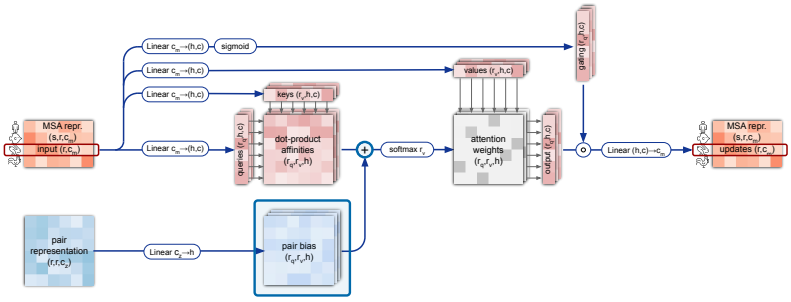
The Evoformer Block

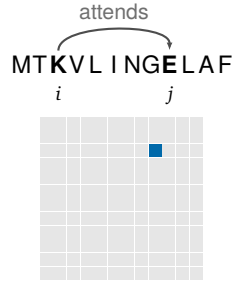


[JEP+21], Fig. 3a

Evoformer: MSA Representation

Self-attention for MSA representation





Residue r_i attends to r_j in its row, biased by the pair b_{ij}.

$$a_{sij} = \text{softmax}_j\left(\frac{1}{\sqrt{c}}\,\mathbf{q}_{si}^{\top}\mathbf{k}_{sj} + b_{ij}\right)$$

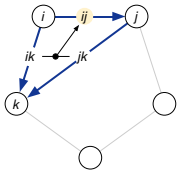
[JEP+21], Suppl. Fig. 2, Algorithm 7

Evoformer: Pair Representation

1. Triangle inequality for the pair representation.

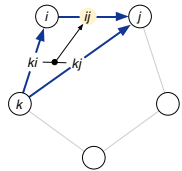
Multiplication

outgoing edge



$\sum_k \mathbf{a}_{ik} \odot \mathbf{b}_{jk}$

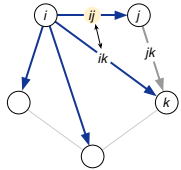
incoming edge



$\sum_k \mathbf{a}_{ki} \odot \mathbf{b}_{kj}$

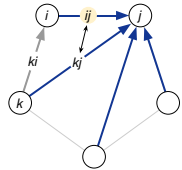
Attention

around starting node



ij from all ik
bias jk

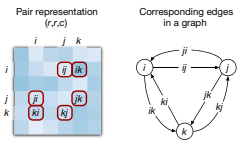
around ending node



ij from all kj
bias ki

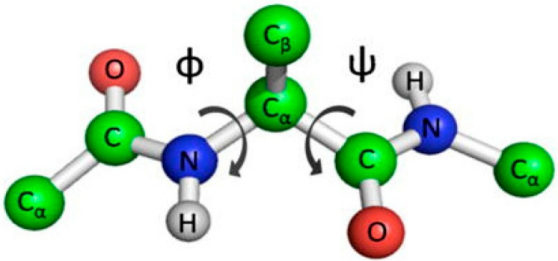
Graph view: residues are nodes, ordered pairs ij are directed edges.

Triangle inequality: $d_{ij} \leq d_{ik} + d_{kj}$ for every third residue k.

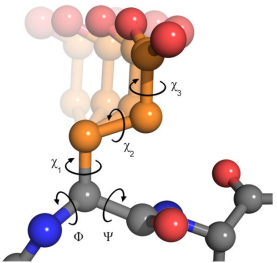


[JEP+21], Fig. 3b and 3c, Supplementary Algorithms 11 to 14

Residue Geometry



Backbone torsion angles ϕ, ψ



Side-chain torsion angles χ_1 to χ_3

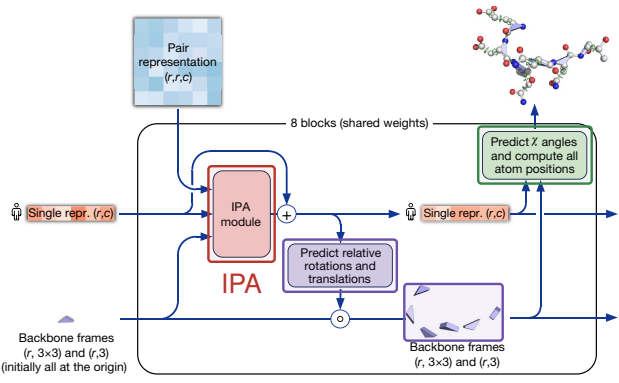
Backbone: N, Cα, C along the chain. *Side chain*: on Cα, 0 to 4 angles χ.

A frame per residue and its torsion angles fix every atom.

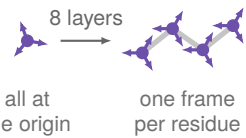
Figures from [RKN+21], Fig. 3a, CC BY 4.0, and [HBP+10], Fig. 1, CC BY 2.0. [JEP+21], SI 1.8.4

The Structure Module

2. Rotation and translation invariance.



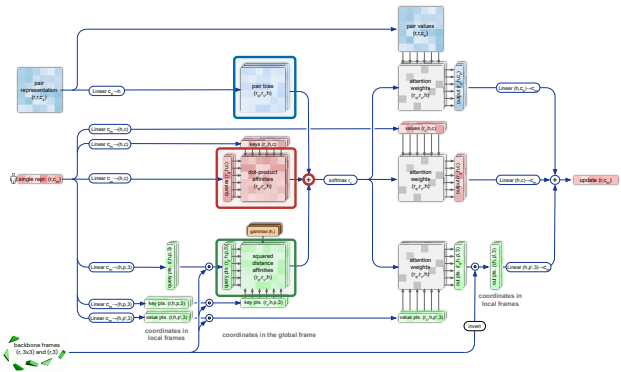
In each of 8 layers *invariant point attention* (IPA) updates the input residue features and position and orientation of each frame. Then, from frames and torsion angles χ, all atom coordinates are computed.



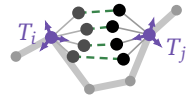
[JEP+21], Fig. 3d

Structure Module: Invariant Point Attention

2. Rotation and translation invariance.



Logit: dot product, plus pair bias, minus the squared distances of the points.



frames T_i, T_j
4 points p per head, in the local frame
distance for each p
protein chain

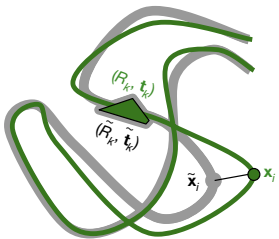
$$\mathbf{q}_i^\top \mathbf{k}_j + b_{ij} - \gamma \sum_p \|T_i \circ \vec{\mathbf{q}}_{ip} - T_j \circ \vec{\mathbf{k}}_{jp}\|^2$$

A global rotation or translation leaves every weight unchanged.

[JEP+21], p. 587. SI 1.8.2, Suppl. Fig. 8, Algorithm 22, one of 12 heads, constant weights omitted. Invariance proof on backup slide I

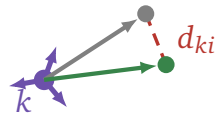
Loss: Frame Aligned Point Error

3. Chirality (mirror images differ).



Frame aligned point error (FAPE): distance d_{ki} between atom i in frame k, predicted and true.

$$\mathcal{L}_{\text{FAPE}} = \frac{1}{10\,\text{Å}} \text{mean}_{k,i} \min(10\,\text{Å}, d_{ki})$$

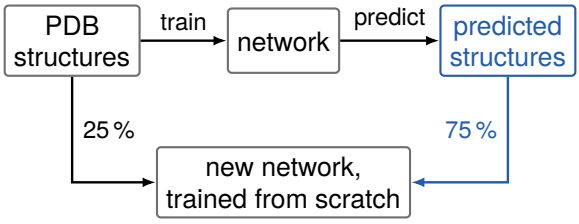


FAPE is invariant to rotations and translations. It distinguishes mirror images.

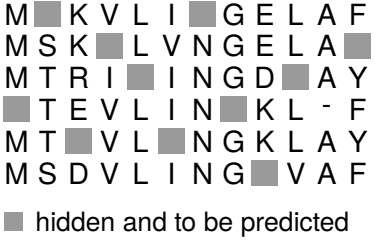
[JEP+21], p. 587 and Fig. 3f. SI 1.9.2, Algorithm 28. Mirror argument on backup slide II

Training Data and Loss

Self-distillation: training on predicted structures for 350,000 sequences without a known structure.



Masked MSA loss: 15 % of entries hidden and predicted, as in BERT.



Total loss: FAPE (final and per layer), distances, masked MSA, pLDDT.

Self-distillation considerably improves the accuracy.

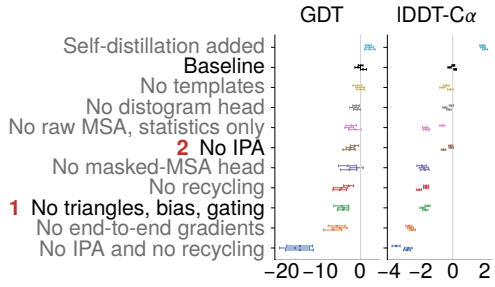
[JEP+21], p. 587. SI 1.3, 1.9, eq. 7, fine-tuning adds two terms. Self-distillation after [XLHL20], masking after [DCLT19]

Ablation Study

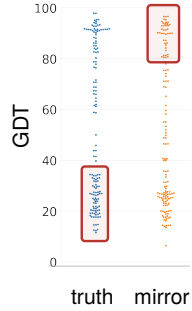
1 2 3

Global distance test (GDT): mean percent of Cα atoms within 1, 2, 4 and 8 Å of the experiment

Accuracy change against the baseline



Without FAPE: GDT against truth and mirror

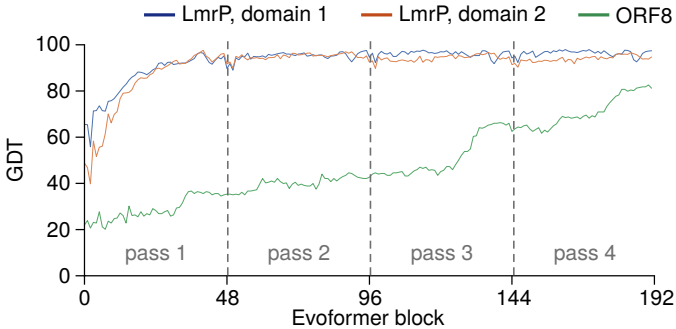


3 Without FAPE, the model frequently predicts mirror images (SI 1.9.3).
Red boxes: low GDT against the truth, high GDT against the mirror.

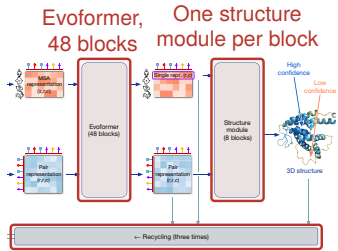
Triangle updates matter more than IPA. FAPE ensures the correct chirality.

[JEP+21], Fig. 4a and Suppl. Fig. 9, labels re-set. GDT (Zemla 2003) on 87 CASP14 domains, lDDT-Cα on PDB chains. SI 1.9.3

Intermediate Structures



From slide 5



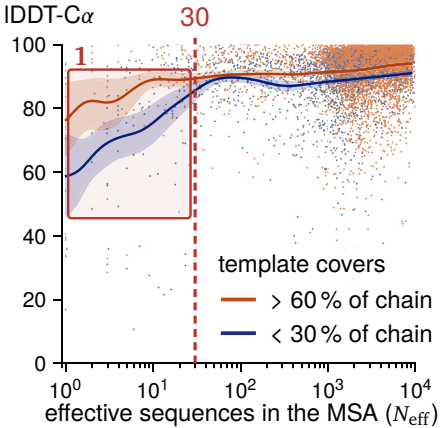
Recycled three times: 4 passes

Probing: each of the 48 Evoformer blocks gets its own copy of the structure module, trained to predict the structure from that block's output. The main network is frozen.

LmrP reaches its final structure early. ORF8 needs nearly the full depth.

[JEP+21], p. 587, Fig. 4b and Fig. 1e, labels re-set, SI 1.14. LmrP: CASP14 target T1024. ORF8: T1064, SARS-CoV-2

Limitations



1 Proteins with few known relatives

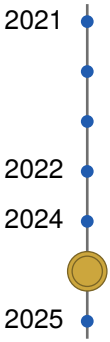
Accuracy drops below about 30 sequences.
With a template, the drop is much smaller.

2 Domains whose shape comes from a complex

Much weaker if most contacts are inter-chain.
AlphaFold 2 takes one chain as input.

[JEP+21], p. 588, Fig. 5a, labels re-set. Recent PDB chains with at most 25 % of contacts to other chains. N_{eff}: sequences after merging near-duplicates

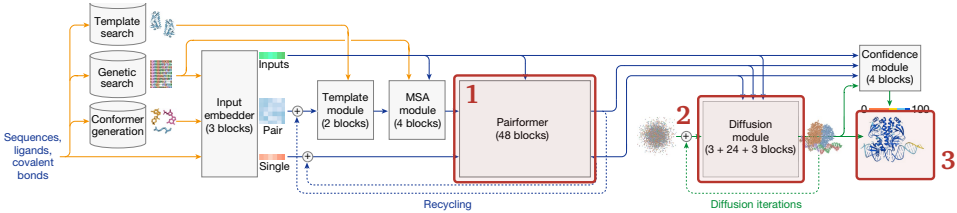
Since AlphaFold 2



2021	RoseTTAFold, published the same day, is nearly as accurate	[BDA+21]
	Structures predicted for 98.5 % of human proteins	[TAW+21]
	AlphaFold-Multimer predicts complexes	[EOP+21]
2022	ESMFold predicts from one sequence alone	[LAR+23]
2024	AlphaFold 3: proteins, DNA, RNA and ligands (small molecules)	[AAD+24]
	Nobel Prize in Chemistry for Baker, Hassabis and Jumper	[Roy24]
2025	AlphaGenome predicts gene regulation from DNA	[ALC+26]

Full references at the end

Design Constraints in AlphaFold 3



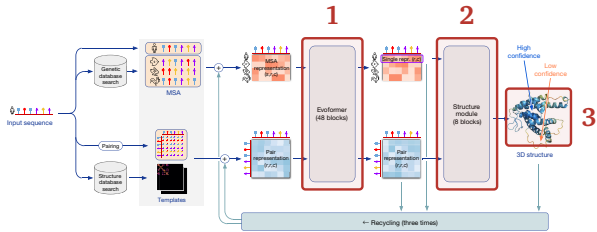
1 The Pairformer keeps the triangle updates.

2 A diffusion module outputs atom coordinates, without frames or IPA.

3 An aligned mean squared error replaces FAPE.
Chirality is wrong in 4.4 % of ligand predictions.

[AAD+24], Fig. 1d, p. 494 to 497, Suppl. 3.7.1. 4.4 %: chirality violation rate on the PoseBusters benchmark, after a ranking penalty

Summary



		Removal in AlphaFold 2	AlphaFold 3
1	triangle updates on the pairs	−4.6 GDT	kept
2	IPA in the structure module	−2.3 GDT	dropped
3	FAPE in local frames	mirror images	dropped, 4.4 % wrong chirality

AlphaFold 2 reaches atomic accuracy in CASP14.
Triangle updates, IPA and FAPE let it learn efficiently from limited data.

AlphaFold 3 also predicts DNA, RNA and ligands, for which AlphaFold 2 defines no frames. Rotations are learned from randomly rotated training data instead.

[JEP+21], p. 584 and 588, Fig. 4a, mean of three seeds. [AAD+24], p. 494 to 497. Figures reproduced under CC BY 4.0

Exam Question

List at least two use cases where AlphaFold will struggle.

1 Proteins with few known relatives

Accuracy drops substantially below a median alignment depth of about 30 effective sequences, as for orphan or designed proteins. A template with high coverage softens the drop.

2 Domains whose shape comes from a complex

Much weaker for chains with few intra-chain contacts relative to contacts with other chains, typically bridging domains in larger complexes. Homomers are often accurate.

[JEP+21], p. 588, Fig. 5a on the limitations slide. Depth: median per-residue N_{eff}, Methods p. 591

Discussion

1 pLDDT is trained to predict the lDDT-Cα of the network's own structure. When would you distrust it?

2 Accuracy drops below a median alignment depth of about 30 effective sequences. Has the network learned the physics of folding, or the statistics of evolution?

3 AlphaFold 2 builds rotation and translation invariance (IPA) and chirality (FAPE) into the network. AlphaFold 3 drops both and keeps the triangle updates. When is a built-in constraint worth it?

`github.com/BenBullinger/ML4Genomics-Presentation-AF2`

References: Structure Prediction

[JEP+21] J. Jumper, R. Evans, A. Pritzel, et al., "Highly accurate protein structure prediction with AlphaFold," *Nature* 596, 583 to 589 (2021). CC BY 4.0.

[TAW+21] K. Tunyasuvunakool, J. Adler, Z. Wu, et al., "Highly accurate protein structure prediction for the human proteome," *Nature* 596, 590 to 596 (2021).

[BDA+21] M. Baek, F. DiMaio, I. Anishchenko, et al., "Accurate prediction of protein structures and interactions using a three-track neural network," *Science* 373, 871 to 876 (2021).

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[LAR+23] Z. Lin, H. Akin, R. Rao, et al., "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science* 379, 1123 to 1130 (2023). Preprint: *bioRxiv* 2022.07.20.500902.

[AAD+24] J. Abramson, J. Adler, J. Dunger, et al., "Accurate structure prediction of biomolecular interactions with AlphaFold 3," *Nature* 630, 493 to 500 (2024). CC BY 4.0.

[MCS+11] D. S. Marks, L. J. Colwell, R. Sheridan, et al., "Protein 3D structure computed from evolutionary sequence variation," *PLoS ONE* 6, e28766 (2011).

References: Methods and Context

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[Sut19] R. Sutton, “The Bitter Lesson,” essay, incompleteideas.net (2019).

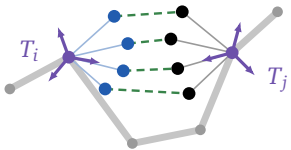
[Roy24] The Royal Swedish Academy of Sciences, “Scientific background to the Nobel Prize in Chemistry 2024” (2024).

[RKN+21] V. R. Rudnev, L. I. Kulikova, K. S. Nikolsky, et al., “Current approaches in supersecondary structures investigation,” *Int. J. Mol. Sci.* 22, 11879 (2021). CC BY 4.0.

[HBP+10] T. Harder, W. Boomsma, M. Paluszewski, et al., “Beyond rotamers: a generative, probabilistic model of side chains in proteins,” *BMC Bioinformatics* 11, 306 (2010). CC BY 2.0.

IPA: Invariance and Equivariance

SI 1.8.2 and 1.8.3, equations 3 to 6



Query-key distances in global coordinates:

$$\left\| T_{\text{global}} \circ T_i \circ \vec{\mathbf{q}}_{ip} - T_{\text{global}} \circ T_j \circ \vec{\mathbf{k}}_{jp} \right\| = \left\| T_i \circ \vec{\mathbf{q}}_{ip} - T_j \circ \vec{\mathbf{k}}_{jp} \right\|$$

Output points in frame i, with $\sum_j a_{ij} = 1$:

$$(T_{\text{global}} \circ T_i)^{-1} \circ \sum_j a_{ij}\, T_{\text{global}} \circ T_j \circ \vec{\mathbf{v}}_{jp} = {T_i}^{-1} \circ \sum_j a_{ij}\, T_j \circ \vec{\mathbf{v}}_{jp}$$

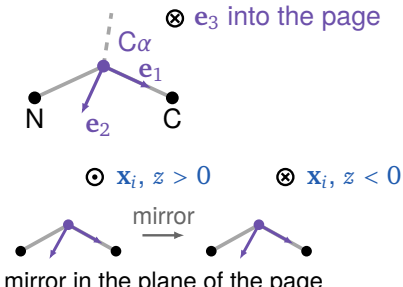
Frame update, composed in local coordinates:

$$(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$$

The attention output is invariant. The frame update is equivariant.

FAPE and Chirality

SI 1.8.1 and 1.9.3, Algorithm 21, equations 11 to 17, Suppl. Fig. 9



Frames from N, Cα, C by Gram–Schmidt.
Always right-handed: $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$.

Mirroring flips the sign of z in each local frame:

$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ki} \left| (T_k{}^{-1} \circ \mathbf{x}_i)_z \right|$$

With a distance-only loss (dRMSD), the model often predicts mirror images.

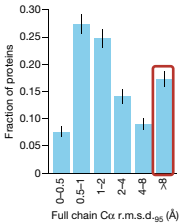
FAPE penalises the mirror image of the true structure.

Generalisation to New Structures

[JEP+21], Fig. 2a and 2b, p. 585

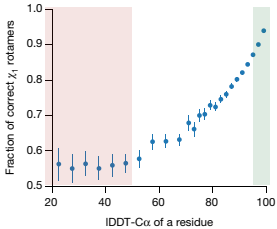
Chains released after training. No template above 40 % identity.

Backbone error per chain



Median 1.46 Å. 17 % lie above 8 Å. Some of these have misplaced domains.

Side chains against backbone



lDDT-Cα near 100: 93 % of side-chain angles χ_1 correct. Below 50: about 55 %.

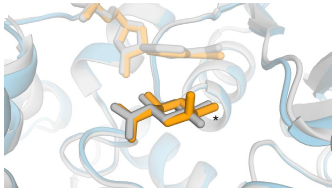
High accuracy on chains released after training.

3,144 and 5,317 chains. Percentages read off the figure. A χ_1 angle counts as correct within 40°

Reported Failures of AlphaFold 3

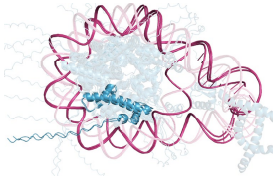
[AAD+24], Fig. 5b and 5e, p. 497 and 499

Wrong chirality



Predicted alpha form, true beta form.
Wrong in 4.4 % of predictions.

Overlapping chains



Protein and DNA overlap.
Mostly in large complexes.

Penalties in sample ranking reduce both failures.

PDB 7CTM and 7PEU. The 4.4 % is on the PoseBusters benchmark